

Yatin Dandi

PH.D STUDENT · SCHOOL OF COMPUTER AND COMMUNICATION SCIENCES

EPFL, Lausanne

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Education

École Polytechnique Fédérale de Lausanne

PH.D, COMPUTER SCIENCE

Lausanne, Switzerland

September, 2022 - Present

- Presently working on Statistical Physics of Computation and Universality under Prof. Lenka Zdeborová and Prof. Florent Krzakala.
- Accepted to the ELLIS program. Main supervisor: Lenka Zdeborová, Secondary (host) supervisor: Bernhard Schölkopf.

Indian Institute of Technology Kanpur

DUAL DEGREE (BTECH-MTECH), COMPUTER SCIENCE AND ENGINEERING

Kanpur, India

July, 2017 - August 2022

- **Cumulative Performance Index : 9.6/10.0, Master's CPI: 10.0/10.0**
- **Exchange and visiting student at Ecole polytechnique fédérale de Lausanne (EPFL)**

Interests

Theory of Deep Learning, Statistical Physics of Computation, Optimization, Causality, Dynamical Systems, Combinatorics.

Publications and Preprints

Single-spin-flip-stability in Random Graphs and Spin Glasses

YATIN DANDI, DAVID GAMARNIK, LENKA ZDEBOROVÁ

In preparation.

Data-heterogeneity-aware Mixing for Decentralized Learning

YATIN DANDI, ANASTASIA KOLOSKOVA, MARTIN JAGGI, SEBASTIAN STICH

NeurIPS 2022 Workshop: Optimization for Machine Learning

New Orleans

2022

Implicit Gradient Alignment in Distributed and Federated Learning

YATIN DANDI*, LUIS BARBA*, MARTIN JAGGI (* DENOTES EQUAL CONTRIBUTION)

Proceedings of the AAAI Conference on Artificial Intelligence (Preliminary version also accepted at the FL-ICML Workshop, 2021)

Virtual Conference

February, 2022

Generalized Adversarially Learned Inference

YATIN DANDI, HOMANGA BHARADHWAJ, ABHISHEK KUMAR, PIYUSH RAI

Proceedings of the AAAI Conference on Artificial Intelligence (Preliminary version also accepted at the NeurIPS 2020 Workshop: Self-Supervised Learning – Theory and Practice)

Virtual Conference

February, 2021

Jointly Trained Image and Video Generation using Residual Vectors

YATIN DANDI, ANIKET DAS, SOUMYE SINGHAL, VINAY P. NAMBOODIRI, PIYUSH RAI

2020 Winter Conference on Applications of Computer Vision (WACV '20)

Colorado, USA

March, 2020

Model-Agnostic Learning to Meta-Learn

ARNOUT DEVOS*, YATIN DANDI* (* DENOTES EQUAL CONTRIBUTION)

Pre-registration workshop, NeurIPS (2020). Full paper published in Proceedings of Machine Learning Research (PMLR).

Virtual Conference

December, 2020

NeurInt-Learning Interpolation by Neural ODEs (Spotlight)

AVINANDAN BOSE*, ANIKET DAS*, YATIN DANDI, PIYUSH RAI

The Symbiosis of Deep Learning and Differential Equations: DLDE Workshop, NeurIPS 2021.

Virtual Conference

December, 2021

Understanding Layer-wise Contributions in Deep Neural Networks through Spectral Analysis

YATIN DANDI, ARTHUR JACOT

Arxiv Preprint: <https://arxiv.org/abs/2111.03972>

Honors & Awards

2017	Aditya Birla Scholarship , Awarded to 15 students from all IITs	Mumbai, India
2018,2019	Academic Excellence Award , Awarded (twice) for exceptional performance	IIT Kanpur
2016	KVPY Scholarship, 2016 , Indian Institute of Science	Bangalore
2015	NTSE Scholarship , Government of India	India
2017	All India Rank 135 , JEE Advanced 2017	India
2016	Selected for Indian National Physics and Chemistry Olympiads , HBCSE	Mumbai, India

Projects and Internships

Single-spin-flip-stability in Random Graphs and Spin Glasses

EPFL

PROJECT UNDER PROF. ZDEBOROVÁ AT EPFL AND PROF. GAMARNIK AT MIT

June 2022 - Present

- Rigorously proved a phase transition characterizing partitions with a large number of vertices having sufficient single-spin-flip-stability or friendliness/unfriendliness in Random Graphs and Spin Glasses.
- Proved universality of the threshold across different distributions of sparse weighted random graphs in the large degree limit.
- Proved that with appropriate re-scaling, the universality result further extends to dense fully-connected spin glass models such as the Sherrington-Kirkpatrick model.
- Utilized a range of theoretical techniques from Combinatorics, Theory of Large Deviations, and Probability Theory.

Optimization, Sampling, and Trajectory Inference algorithms

Montreal Institute for Learning Algorithms

INTERNSHIP UNDER PROF. SIMON LACOSTE-JULIEN AND PROF. YOSHUA BENGIO

June 2022 - August 2022

- Worked on the analysis of Saddle Point Frank Wolfe algorithm.
- Worked on Sampling and Trajectory Inference Algorithms motivated by applications in single-cell Biology

Data dependent mixing for Faster Heterogeneous Decentralized Learning

EPFL

PROJECT UNDER PROF. JAGGI AT THE MLO LAB, EPFL

August 2021 - Present

- Extended the analysis for decentralized SGD to characterize the interaction between data heterogeneity and the mixing matrix.
- Formulated a novel criterion for periodically updating the mixing matrix based on an appropriate constrained optimization problem.
- Implemented the approach in toy settings as well as on large scale datasets using the torch.distributed and MPI framework.

Implicit Gradient Alignment in Distributed and Federated Learning

EPFL

PROJECT UNDER PROF. JAGGI AT THE MLO LAB, EPFL

April 2021 - August 2021

- Studied several theoretical works aiming to explain the generalization benefits of SGD.
- Designed a novel algorithm that replicates the implicit regularization effect of SGD in distributed and federated learning settings.
- Proved theoretical results that show that the proposed algorithm approximates SGD and leads to descent on the regularized objective.
- Demonstrated an analogous implicit regularization effect for SCAFFOLD.

Generalized Adversarially Learned Inference

IIT Kanpur

UNDERGRADUATE PROJECT UNDER PROF. PIYUSH RAI

October 2019 - May 2020

- Proposed a novel approach to introduce reconstructive, self-supervised, and knowledge-based feedback in the ALIGAN/BIGAN framework by modifying the discriminator's objective to correctly identify more than just two joint distributions of image-latent vector pairs.
- Performed quantitative and qualitative evaluations over baselines on SVHN and CelebA datasets.
- Proved that the optimum for the proposed objective corresponds to simultaneously matching all the distributions.

Understanding Layer-wise Contributions in Deep Neural Networks through Spectral Analysis

EPFL

PROJECT UNDER ARTHUR JACOT AND PROF. CLEMENT HONGLER AT EPFL

March 2020 - May 2021

- Utilized the properties of Hermite Polynomials and Spherical Harmonics to prove that the eigenvalues corresponding to high frequency functions of the contribution to the NTK from the first layer are larger than those corresponding to the second layers for a two-layer network with arbitrary activation functions.
- Validated the proposed theory using experiments on datasets of Spherical Harmonics as well as MNIST.

Causal Inference, Out of Distribution Generalization and Meta Learning: A Unified Perspective

EPFL

SEMESTER PROJECT UNDER PROF. GROSSGLAUSER AT THE INDY LAB, EPFL

August 2020 - December 2020

- Studied the theory of causal inference, surveyed recent works on causal inference and out of distribution generalization for several classes of machine learning models
- Analyzed the relationship between out of distribution generalization, meta-learning and causal inference
- Formulated and empirically analyzed several algorithms under the novel paradigm of "out of task distribution generalization".

On the Effect of Noise induced by Gradient Stochasticity on Optimizing 2-player Differentiable Games

EPFL

PROJECT UNDER TATJANA CHAVDAROVA AT EPFL

January 2021 - May 2021

- Analyzed the continuous time limit of stochastic variants of first order algorithms for differentiable games using the theory of Stochastic Differential Equations (SDEs)
- Derived and studied first order approximations for second order algorithms such as SGA (Symplectic Gradient Adjustment) for n-player games.

Towards a Proof of the Fourier Entropy Conjecture

IIT Kanpur

COURSE PROJECT UNDER PROF. RAJAT MITTAL

August 2021 - November 2021

- Surveyed several papers related to the Fourier Entropy conjecture.
- Prepared a summary report and presentation on the recent work by Kelman et al..

Jointly Trained Image and Video Generation using Residual Vectors

IIT Kanpur

UNDERGRADUATE PROJECT UNDER PROF. PIYUSH RAI AND PROF. VINAY NAMBOODIRI

May 2019 - July 2019

- Designed a novel modelling technique for joint image and video generation that simultaneously learns to map latent variables with a fixed prior onto real images and interpolate over images to generate videos.
- Implemented Variational Autoencoder and Generative Adversarial Networks based video generation models in Pytorch utilizing the proposed technique.
- Performed quantitative and qualitative evaluations over previous VAE and GAN based video generation approaches.

New York Office, IIT Kanpur

IIT Kanpur

MACHINE LEARNING FOR LARGE SCALE LOGISTICS PLATFORM, UNDER PROF. MANINDRA AGARWAL

May 2018 - July 2018

- Implemented a state of the art algorithm for online collaborative filtering based on Fast Matrix Factorization for Online Recommendation with Implicit Feedback (He et al.) using Numpy and improved the model with sentiment and frequency dependent weighting schemes.
- Used Kafka for real-time data processing and simulated interactions using locust.
- Implemented a recommender system based on deep autoencoders and compared the results with other models using metrics such as hit ratio.
- Implemented a Bidirectional LSTM model using Keras for sentiment analysis of user comments.
- Trained the Latent Dirichlet allocation model on Wikipedia articles for automatic extraction of topics.

Social Situation Inference in Simple Animated Shapes

IIT Kanpur

UNDERGRADUATE PROJECT UNDER PROF. NISHEETH SRIVASTAVA

May 2018 - July 2018

- Conducted experiments to characterise the nature and granularity of information about social interactions accessible to human observers from simple visual displays.
- Developed an animation engine capable of producing a vast range of social situations involving autonomous agents having long term and short term goals.

Deep Reinforcement Learning for Atari Games

IIT Kanpur

ASSOCIATION FOR COMPUTING ACTIVITIES, IIT KANPUR

February 2018 - May 2018

- Used Numpy to implement various reinforcement learning algorithms such as Dynamic Programming (Policy and Value iteration), Monte Carlo (Epsilon-greedy and off-policy), TD Learning (Q-Learning and SARSA) and Q-Learning with Function Approximation.
- Implemented Deep Q-Learning and Policy Gradient methods for Atari Games using PyTorch and OpenAI Gym.

Image Captioning with Visual Attention

IIT Kanpur

PROGRAMMING CLUB, IIT KANPUR

February 2018 - May 2018

- Compared various CNN based models for image classification and implemented them using eager execution in Tensorflow.
- Implemented the model described in Show, Attend and Tell (Xu et al.2015) using Tensorflow's estimator API and evaluated the model on MS COCO dataset.

Microsoft code.fun.do hackathon

IIT Kanpur

NATIONAL-LEVEL HACKATHON WINNER

February 2018

- Selected out of 120 students to represent IIT Kanpur at Microsoft Hyderabad center to showcase our project in their academia-industry collaboration event AXLE.
- Built an interactive interface using D3.js to display changing geopolitical relations and popularity of world leaders.
- Used Scrapy, a scraping framework to extract world news.
- Performed sentiment analysis and named entity-recognition on the extracted news to infer the effect of the concerned news on world politics.

Positions of Responsibility

AISTATS, 2022

online

REVIEWER

October 2021 - November 2021

- Reviewed papers on Random Feedback and Adversarial Risk.

Introduction to Machine Learning

IIT Kanpur

TEACHING ASSISTANT

August 2021 - December 2021

- Graded assignments and exams for the introductory machine learning course.

Mathematics for Computer Science - 2, 3

IIT Kanpur

TEACHING ASSISTANT

January 2022 - Present

- Graded assignments and exams for courses on "Logic for Computer Science" and "Probability for Computer Science"

SIGML - Special Interest Group on Machine Learning

IIT Kanpur

COORDINATOR

June 2019 - Present

- Helped conduct and organize talks and lectures on various topics in machine learning.

Programming Club

SECRETARY

- Helped conduct and organize Linux fest, introductory workshops and various hackathons.

IIT Kanpur

April 2018 - April 2019

Association of Computing Activities

PROJECT MENTOR

- Mentored first-year students on a semester long project on Generative Adversarial Networks.

IIT Kanpur

January 2019 - April 2019

Programming Club

PROJECT MENTOR

- Mentored first-year students on a semester long project on Probabilistic Machine Learning.

IIT Kanpur

May 2019 - July 2019

Relevant Courses

Introduction to Programming - A	Probability for Computer Science - A*	Calculus and Real Analysis - A*
Discrete Mathematics - A	Logic for Computer Science - A*	Topics in Probabilistic Modeling and Inference - A
Computational Cognitive Science (audited)	Linear Algebra and ODE - A	Data Structures and Algorithms - A
Bayesian Analysis - A*	Theory of Computation - A	Introduction to Machine Learning - A*
Optimization For Machine Learning - 6.0/6.0	Learning Theory - 6.0/6.0	Statistical Physics of Computation - 6.0/6.0
Mathematical Methods For Physicists	Computational Complexity - A	Complexity Measures for Boolean Functions - A
Quantum Computing - A	Modern Cryptology - A	Lattice Models

A*: grade for exceptional performance

Programming Skills

Programming Languages	C, C++, Python, Javascript, MATLAB, Haskell
Libraries and frameworks	Tensorflow, Pytorch, Jax, Scikit-Learn, Pillow, Keras, Numpy
Web	Flask, HTML, CSS, jQuery, MySQL
Utilities	Linux shell utilities, LaTeX, Git, Docker

Miscellaneous

- Delivered a talk on Zero Knowledge Proofs for the course Complexity Theory, IIT Kanpur and on its applications to machine learning at the MLO lab, EPFL.
- Delivered a talk on Brain-Computer Interface in an event organized by Science Coffee House, IIT Kanpur.
- Winner of Blackbox - a three hour high speed hackathon based on an Esoteric language organized by Programming Club IIT Kanpur.